Marine versus Automotive Alternators

Before one of your dockmate electrical "experts" (there's one in every marina) tells you otherwise, remember that there is a big difference between marine and automotive alternators. The brushes are sealed on ignition-protected marine alternators used on gasoline engines; this prevents dangerous sparks that could cause an explosion. (If you want to see the difference, just open the hood of the family car in the dark while it's running. You're likely to see sparks arcing inside the ventilation ports of the alternator.) Other differences between alternators for cars and those for boats are that marine alternators are of heavier construction, may have a higher output, and have fewer corrosive parts than those for cars do. Don't let anyone talk you into substituting. This is an area mandated by the USCG on gasoline-powered vessels.

Those ventilation ports mentioned above also testify to an alternator's need for cooling air. That thing on the pulley that looks like a fan is, in fact, a fan. Often these fans have blades that are set to work in one direction only, and the direction of engine rotation must be considered. If you have a twin-engine boat, the engines may rotate in opposite directions to compensate for torque. The alternators on twin-engine boats may not be interchangeable without switching the pulleys and fans. Straight fan blades on an alternator usually indicate that the unit will cool rotating in either direction. If the blades are offset, look for an arrow stamped on one of the blades that will show the direction of rotation, and always check the manufacturer's literature for exact installation instructions.

Alternator Electrical Connections

The best bet for identifying the wires connected to your alternator is to use your engine's workshop manual and wiring diagram. If, after testing your alternator, you discover that it must be removed for overhaul or replacement, carefully mark all the wires with tape. Also mark the terminal identification on

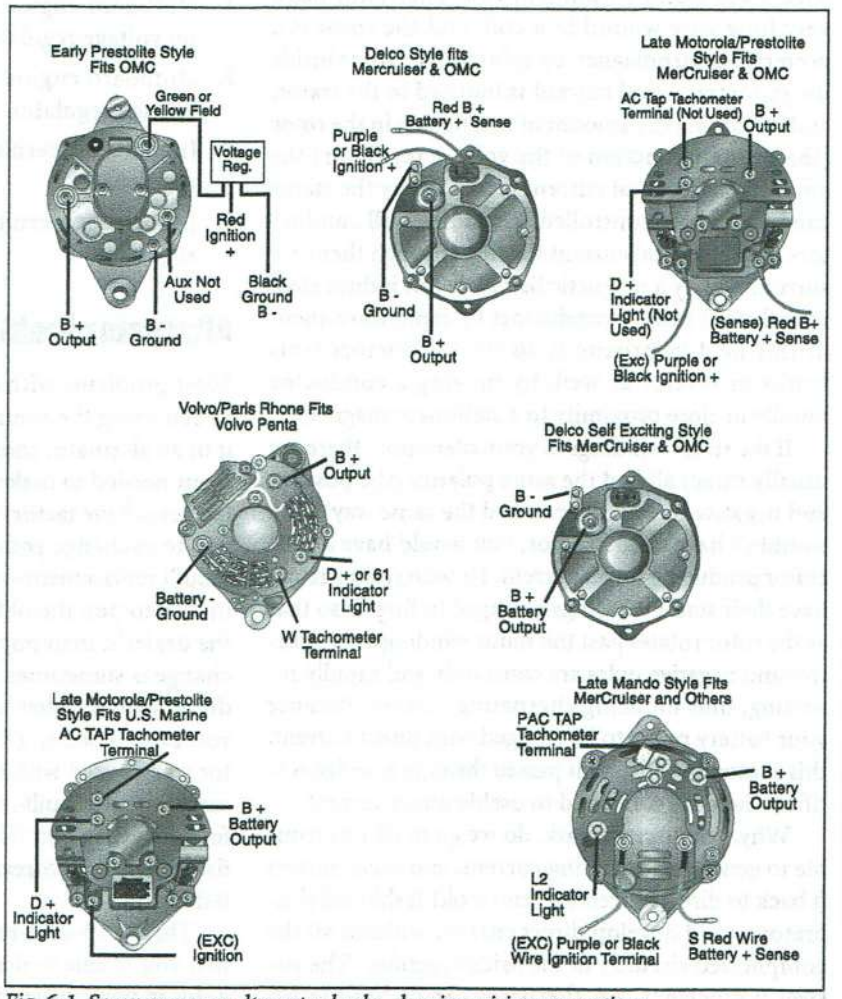


Fig. 6-1. Seven common alternator backs, showing wiring connections.